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Sri Adichunchanagiri Shikshana Trust (R.)

SJC INSTITUTE OF TECHNOLOGY

An Autonomous Institution under VTU from 2024-25

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A National Level Intercollegiate 24-Hour Hackathon



14th & 15th May 2026

Theme-1: Artificial Intelligence for Social Good

Sub theme under Artificial Intelligence for Social Good

- A. AI-Powered Healthcare diagnostics
- B. Smart agriculture using ML
- C. Disaster prediction & response systems
- D. Personalized education platforms

A. AI-Powered Healthcare diagnostics

1. AI-based medical imaging (X-ray, CT, MRI analysis)

Problem Statement:

Develop an AI-driven diagnostic system that can analyze medical images such as rays CT Scan and MRI scans to accurately detect abnormalities (e.g., tumors, fractures, infections) at an early stage, assisting radiologists in reducing diagnostic errors and improving speed.

2. Predictive analytics for disease outbreaks

Problem Statement:

Design a predictive analytics platform using AI/ML techniques that analyzes historical health data, environmental factors, and real-time inputs to forecast potential disease outbreaks, enabling early warning and proactive public health response.

3. Wearable health monitoring systems

Problem Statement:

Create an intelligent wearable health monitoring system that continuously tracks vital parameters (such as heart rate, SpO₂, and activity levels) and uses AI algorithms to detect anomalies and provide real-time alerts for preventive

healthcare.

4. AI chatbots for primary healthcare support

Problem Statement:

Develop an AI-powered chatbot capable of providing preliminary medical guidance by analyzing user-reported symptoms, offering basic health advice, and directing patients to appropriate healthcare services while ensuring accuracy and user privacy.

B. Smart agriculture using ML

1. Crop Disease Detection using Image Processing

Problem Statement:

Develop an AI-based image processing system that can identify and classify crop diseases from leaf images captured via smartphones or drones, enabling farmers to take timely and accurate corrective measures to reduce crop loss.

2. Soil Quality and Moisture Prediction

Problem Statement:

Design a machine learning model that predicts soil quality and moisture levels using sensor data and environmental parameters, helping farmers optimize crop selection and irrigation planning for improved productivity.

3. Smart Irrigation Systems

Problem Statement:

Create an intelligent irrigation system that uses real-time data (soil moisture, weather conditions, crop type) and AI algorithms to automate water supply, minimizing water wastage while ensuring optimal crop growth.

4. Yield Prediction Models

Problem Statement:

Develop a predictive model that estimates crop yield based on historical data, weather patterns, soil conditions, and farming practices, enabling farmers and policymakers to make informed decisions on production and supply chain management.

C. Disaster prediction & response systems

1. Flood, Earthquake, and Cyclone Prediction Models

Problem Statement:

Develop an AI-based prediction system that analyzes historical data, satellite imagery, and real-time environmental parameters to forecast natural disasters such as floods, earthquakes, and cyclones with improved accuracy and early warning capabilities.

2. Real-time Disaster Monitoring Systems

Problem Statement:

Design a real-time disaster monitoring platform that integrates data from IoT sensors, satellites, and weather stations to detect and track disasters as they occur, providing timely alerts and situational awareness to authorities and communities.

3. AI-based Rescue Route Optimization

Problem Statement:

Create an AI-driven system that determines the safest and fastest routes for rescue teams and evacuation during disasters by analyzing real-time traffic, road conditions, and hazard zones to minimize response time and save lives.

4. Social Media Data Analysis for Emergency Alerts

Problem Statement:

Develop an AI-based system that uses natural language processing (NLP) to analyze social media data in real time, identify disaster-related posts, and generate early alerts and actionable insights for emergency response agencies.

D. Personalized Education Platforms

1. Adaptive Learning Systems Based on Student Performance

Problem Statement:

Develop an AI-driven adaptive learning system that continuously analyzes student performance, learning pace, and weaknesses to dynamically adjust content difficulty and provide personalized learning paths for improved academic outcomes.

2. AI Tutors and Chatbot-Based Learning Assistants

Problem Statement:

Create an AI-powered virtual tutor or chatbot that can interact with students in real time, answer academic queries, explain concepts in multiple ways, and provide guidance similar to a human tutor while being accessible anytime.

3. Learning Analytics and Performance Prediction

Problem Statement:

Design a learning analytics system that uses machine learning techniques to analyze student data (attendance, assessments, engagement) and predict academic performance, helping educators identify at-risk students and take timely interventions.

4. Content Recommendation Systems

Problem Statement:

Develop an AI-based recommendation system that suggests relevant learning resources (videos, notes, quizzes) tailored to individual student preferences, learning history, and performance to enhance engagement and knowledge retention.

Theme-2: Sustainable Mobility for the Future

Sub theme under Sustainable Mobility for the Future

- A. Hypersonic vehicle Aerodynamic design
- B. Navigate, Adapt & Survive in Adverse Flight Conditions

A. Hypersonic Vehicle Aerodynamic Design

1. Computational Fluid Dynamics (CFD) for High-Speed Flows

Problem Statement:

Develop a CFD-based simulation model to analyze high-speed (hypersonic) airflow around a vehicle, capturing shock waves, boundary layer effects, and heat transfer, to optimize aerodynamic performance and minimize drag under extreme flight conditions.

2. Thermal Protection Systems (TPS)

Problem Statement:

Design an efficient thermal protection system capable of withstanding extreme aerodynamic heating during hypersonic flight by selecting suitable materials and cooling techniques, ensuring structural integrity and passenger/payload safety.

B. Navigate, Adapt & Survive in Adverse Flight Conditions

1. Adaptive Flight Trajectory Management under Dynamic Weather Constraints

Problem Statement

Develop a system capable of dynamically adapting aircraft flight trajectories under adverse atmospheric conditions such as turbulence, wind disturbances, and storm regions while maintaining operational safety, maneuverability limits, and energy-efficient flight performance.

2. Intelligent Turbulence and Icing Awareness System for Sustainable Flight Operations

Problem Statement

Design a computational framework to assess and recommend safe flight paths for aircraft operating in adverse flight conditions by analyzing atmospheric disturbances, operational constraints, and aircraft maneuverability in real time.